ABISHEK THIRUNAVUKKARASU

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PROFESSIONAL SUMMARY

Engineering graduate from SASTRA University with hands-on experience building and deploying AI-powered solutions. Designed and implemented an autonomous AI voice assistant that orchestrates OpenAI's GPT API, speech-to-text engines, and computer vision into a seamless conversational experience. Passionate about integrating APIs, troubleshooting complex technical setups, and turning AI technology into practical tools that deliver real value.

EDUCATION

B.Tech in Electronics and Communication Engineering	SASTRA University - CGPA 6.5081	**2021 - 2026**
Higher Secondary Certificate - HSC Grade 12	Don Bosco School - 87%	**2021**

TECHNICAL SKILLS

AI & Automation : OpenAI API, Conversational AI, TensorFlow, Deep Learning, Chatbot Development

APIs & Integration : REST APIs, API Orchestration, SaaS Platforms, Workflow Automation

Programming : Python, Java, SQL

Software Engineering: Object-Oriented Programming (OOP), Exception Handling, File I/O, Serialization

Tools & Platforms : Git, Linux, Streamlit, GitHub Actions, Raspberry Pi

Soft Skills : Technical Troubleshooting, Problem Solving, Client Communication, Technical Documentation, Cross-Functional Collaboration

PROJECTS

Cognitive Human Interactive Technological Intelligence (CHITI). *link to github* 2026

- Designed and built an end-to-end AI agent that integrates OpenAI GPT-4o mini API, Sarvam AI speech-to-text, and Piper TTS into a unified conversational workflow (Wait → Listen → Think → Speak).
- Implemented real-time face tracking using OpenCV and coordinated multi-platform API calls to enable features like live news retrieval, email management, and web search.
- Trained a custom wake word detection model using Open Wake Word to trigger the assistant hands-free.
- Built the complete system from concept to deployment on Raspberry Pi 5, managing hardware-software integration across multiple subsystems.

Early Diagnosis of Autoimmune Skin Conditions using Neural Networks. *link to github* 2025

- Trained and compared 6 deep learning architectures to classify 5 autoimmune skin conditions, achieving 81% accuracy with EfficientNet-B0.
- Deployed the model as a user-facing Streamlit web application for clinical screening turning a complex AI model into an accessible tool.
- Built end-to-end data pipelines for image preprocessing, augmentation, and model optimization using Python, TensorFlow, and NumPy.

Flight Computer for Model Scale Rockets. *link to github* 2024

- Designed a real-time data pipeline that reads 6-axis sensor data, applies Kalman filtering, and logs telemetry all within strict timing constraints.
- Designed and implemented a closed-loop control algorithm to dynamically adjust flight parameters based on continuous real-time data feedback.
- Built a companion mobile app with REST APIs for remote monitoring and wireless configuration of the system.
- Demonstrated strong skills in system integration, debugging under pressure, and end-to-end solution delivery.

PORTFOLIO

Designed and deployed a custom portfolio website from scratch to effectively highlight professional milestones and project portfolios.Wrote clean, semantic HTML and modular CSS to ensure cross-browser compatibility and strict adherence to web accessibility standards. Click here : abishek.in